

[1 hour]

Introduction

A pattern is an arrangement of descriptors. Pattern is something that is set up as an ideal to be followed.

Example: A ~~set~~ shape "A" is shown to a child & asked to copy that. Hence, that shape is ideal. The child draws the shape close to that shape.

His teacher identifies as a 'A'. This step is known as recognition.

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Pattern Recognition

It is defined as the classification of input data through the extraction of important features from a lot of ~~in~~ noisy data.

Example: PR involves the identification of faces, objects, words and melodies, etc.

Pattern vectors are represented by following

representation method:

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Where,

n = total no. of descriptors

X = X -descriptors

$x_1, x_2, \dots, x_n$ = each component of descriptors

Applications of Pattern Recognition

PR has been applied in many fields such as Artificial Intelligence, Computer Engg., medicine, image analysis, etc.

Some major applications of PR are as follows:-

(i) Computer Aided Diagnosis (CAD)

It helps in medical imaging such as ~~EEG~~ ECG, EEG, X-Ray, etc.

(ii) Handwriting Recognition

PR helps in handwriting recognition such as processing bank cheque and pen based input on PR.

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(iii) Speech Recognition

It is also used in speech recognition such as speech signal is classified into phonemes and words.

(iv) Safety (Biometric Identification)

PR also used for biometric identification such as face, fingerprint & iris recognition.

(v) Astronomy

It is also used in astronomical fields such as classifying the galaxy by shape & telescope.

(vi) Agriculture

It is also used in agricultural fields such as soil evaluation and extraction of minerals.

(vii) Geography

It is used for rock classification & identification is geography.

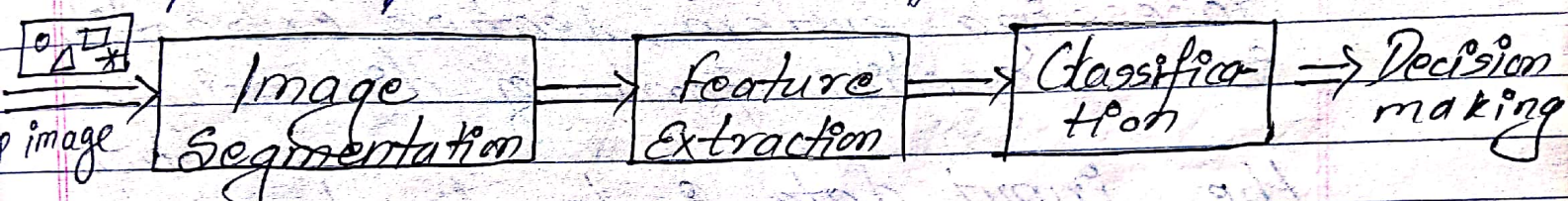
(viii) Engineering

It can be used for diagnosis of vehicle system and recognition of automobile.

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Components of PR

Steps or Phase of PR



(fig: 3-phase of PR)

The phases for pattern recognition can be summarized as follows:

(i) Image Segmentation

The first phase of pattern recognition is segmentation in which the image is isolated from the rest of image.

Segmentation refers to the process of partitioning a digital image into multiple regions (set of pixels).

So, the goal of segmentation is to simplify and change the representation of an image into

meaningful information & easier to analyse.

It is used to locate object and boundaries.

(ii) Feature Extraction

In PR and Image processing system, feature extraction is a special form of dimensioning.

When the input data is too ~~big~~ large to be processed and is ~~it~~ suspected to be redundant, then the input data will be transformed into a reduced & representable set of features.

Thus, transforming the input data into the set of features is known as feature extraction.

Hence, objective of feature extraction is data reduction by transformation method with a certain features.

Example: Spatial features, transform features, edge & boundaries features, shape features, moment features, texture features, color features etc.

(iii) Classification

It is the mechanism to classify pattern according to the extracted features.

It is based on the availability of the set of pattern that have already been classified which is termed as training set.

(Design Cycle of PR)

[Data Collection]

[feature selection]

[Classifier design]

[System evaluation]

(Output)

(fig: Classification Process)

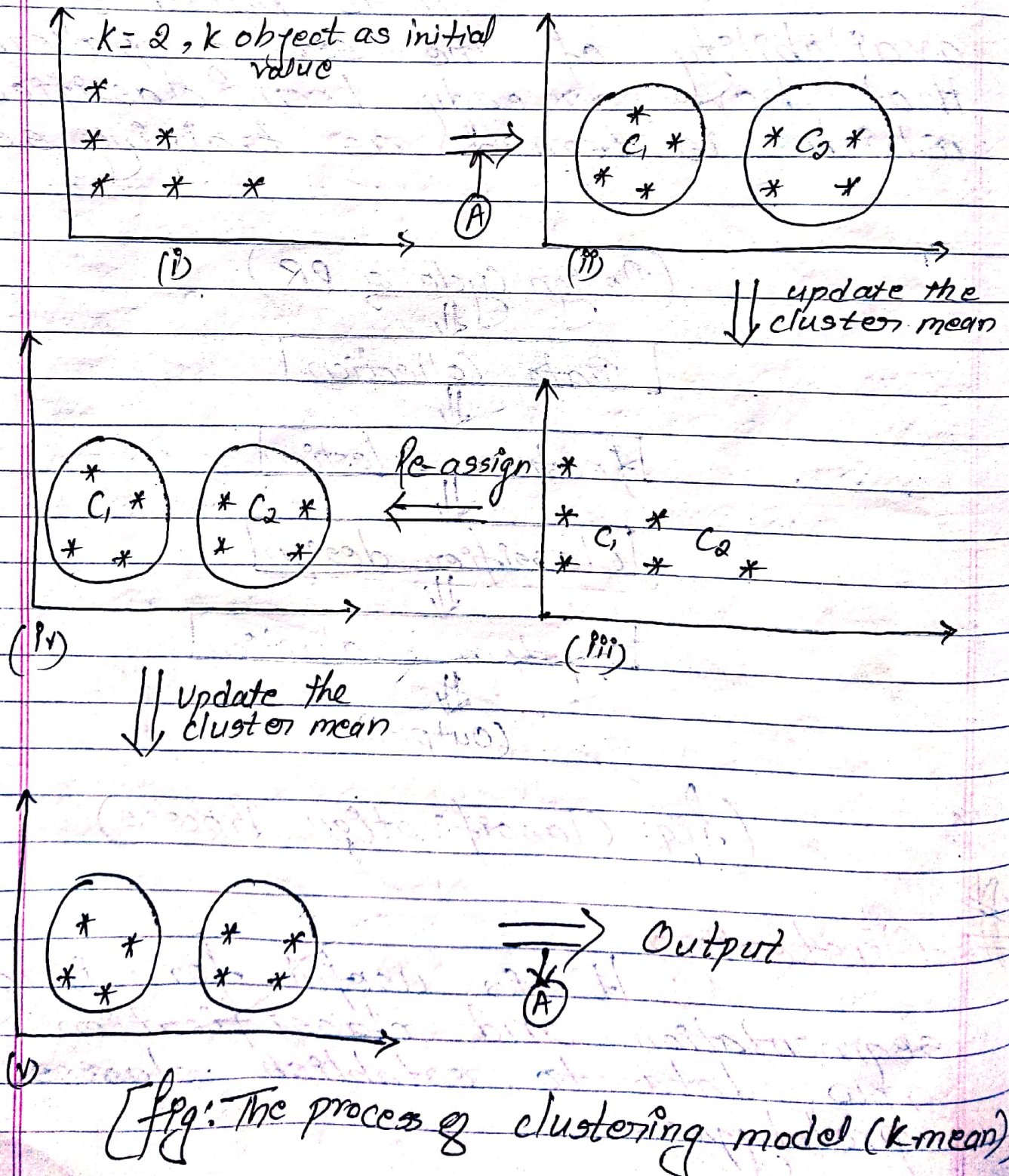
Clustering

It is useful for image segmentation and classification of raw data to establish classes and prototypes.

→ Clustering is the task of grouping a set of objects in such a way that objects in the same group (called a cluster) are more similar (in some sense) to each other than to those in other groups (clusters).

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One of the most popular clustering technique is K-mean algorithm. Clustering is implemented as:



Extra materialDifferences between
DIP

DIP	CG (Computer Graphics)
(i) Emphasis is on analysis and enhancement of image.	(i) Emphasis is on synthesizing new image from geometry, lighting, parameters, materials & texture.
(ii) It is a vector and raster technique.	(ii) It is a raster technique.
(iii) Example: zooming & histogram technique.	(iii) Example: 2D & 3D paging technique.